

	Project Name:	2.5 GW Cell Line Manufacturing Project, Narmadapuram, MP	Document No.:	01260944_DOC_M_007
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8.12 FLEXIBLE DUCT:

Insulated flexible duct should be UL 181 CLASS I AIR DUCT LISTED AND LABELLED WITH NFPA 90A & 90B AND SEAL OF AIR DIFFUSION COUNCIL with double lamination of tough polyester which encapsulates steel helix wire forms the air tight inner core, double layer core wrapped in a multiple thickness of fiberglass wool with R Value 4.2, Green guard certification of fiberglass wool must. , Reinforced and sheathed in a rugged and durable tridirectionally reinforced metalized polyester jacket.

Flexible duct connections should be made as per UL181 listing procedure with proper flexible right forming brace connection allowing right connections for flexible duct into energy efficient. and strapping the flexible duct connections with flexible duct strap ties.

8.13 TESTING AND BALANCING

After the installation of the entire air distribution system is completed in all respects, all ducts shall be tested for air leaks by visual inspection.

The entire air distribution system shall be balanced using an anemometer. Measured air quantities at fan discharge and at various outlets shall be identical to or less/excess than 5 percent in excess of those specified and quoted. Branch duct adjustments shall be permanently marked after air balancing is completed so that these can be restored to their correct position if disturbed at any time. Complete air balance report shall be submitted for scrutiny and approval, and four copies of the approved balance report shall be provided with completion documents.

9. INSULATION

9.1 SCOPE

The scope of this section comprises the supply and application of insulation conforming to these specifications.

9.1.1 THERMAL INSULATION WORKS (HVAC Duct Insulation) - Indoors

- Insulation material shall be Closed Cell Elastomeric Nitrile Rubber factory laminated with 12 microns glass reinforced aluminium foil with 3x3 pattern.
- Insulation material shall be closed cell flexible elastomeric foam based on synthetic rubber composition with fire retardant properties.
- Thermal conductivity of insulation material shall not exceed 0.036 W/(m·K) at mean temperature of 24°C as per ASTM C 518 / ASTM C 177.
- The insulation material shall have fire performance rating of 25/50 as per ASTM E 84.
- The insulation material shall meet ASHRAE 90.1, NFPA 90A & NFPA 90B Standards.
- The insulation material shall give enhanced level of protection against harmful Microbes such as fungi and mold and should conform to ASTM G 21, UL 181 & ASTM E 2180.
- Water Vapor Permeability of insulation material shall be minimum 0.05 Perm-inch. without any external barrier as per ASTM E 96 Procedure A.
- Insulation to be FM approved as per FM 4924.
- Insulation to be approved by local fire department.
- Moisture Diffusion Resistance Factor or 'μ' value of insulation material shall be minimum 12,000 per ASTM-E96
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- The insulation material shall have water absorption of 0.2% by volume as per ASTM C 1763.

Comply with
ASTM E 84

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- The insulation material shall have fire performance of V0, HB as per UL 94, 1996.
- The insulation material shall be self-extinguishing, does not drip, and does not spread flames.
- The insulation material shall be dust and fiber free.
- The insulation material shall withstand maximum surface temperature of +105 Deg. C and minimum surface temperature of -50 Deg. C as per ASTM C 411 and ASTM C 534.
- The insulation material shall have anti-microbial product, which is EPA (Environmental Protection Agency) Approved, to be used in Duct works and should be an integral part of insulation that cannot be washed off or worn off and.
- The insulation material shall have good ozone resistance as per ASTM D 1149.

Comply

9.1.2 THERMAL INSULATION WORKS (HVAC Duct Insulation) – Outdoors/Shafts/Terrace

- Insulation material shall be Closed Cell Elastomeric Nitrile Rubber
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m·K) at mean temperature of 0°C as per EN 12667.
- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- The insulation material shall have fire performance of V0 & HB as per UL 94, 1996.
- Unlike other generics it should not drip or propagate fire and should be self-extinguishing
- Density of insulation material shall be between 40 to 55 Kg/m3.
- The insulation material shall be dust and fiber free.
- The insulation material shall withstand maximum surface temperature of +85 Deg. C and minimum surface temperature of 0 Deg. C as per EN 14706.
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.

Comply

Outer Covering –

- Insulation Covering material shall be of multiple layered laminate of polymeric material reinforced with scrim with a special UV protection.
- Covering material shall be non-metallic.
- Covering material shall provide mechanical resistance with an excellent aesthetic look.
- Covering material shall be mat finish.
- Covering material shall be of 380 ± 20 GSM as per ASTM D 646.
- Covering material shall be of 380 – 400 microns as per ASTM E 252
- Covering material shall have the tensile strength of >5 MPa in MD & CD as per ASTM D 638
- Covering material shall have good puncture and tear resistance.
- Covering material shall have good UV resistance as per ISO 4892-2 Method A tested for 1000 Hrs.

Noted

9.1.3 THERMAL INSULATION WORKS (Chilled Water / Process Cooling tower / Hot Water)

- Insulation material shall be Closed Cell Elastomeric Nitrile Rubber
- Thermal conductivity of insulation material shall not exceed 0.035 W/(m·K) at mean temperature of 0°C as per EN 12667.

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Comply

- The insulation material shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame and passes Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- The insulation material shall have fire performance of V0, & HB as per UL 94, 1996.
- Unlike other generics it should not drip or propagate fire and should be self-extinguishing
- Density of insulation material shall be between 40 to 55 Kg/m3.
- The insulation material shall be dust and fiber free.
- The insulation material shall withstand maximum surface temperature of +85 Deg. C and minimum surface temperature of -50 Deg. C as per EN 14706.
- The insulation material shall be ROHS (Resistance of Hazardous Substance) compliant.
- The material shall have ODP (Ozone Depletion Potential) and GWP (Global Warming Potential) of Zero.
- Thickness of the insulation shall be as specified for the individual application.

Outer Covering –

Noted

- Insulation Covering material shall be of multiple layered laminate of polymeric material reinforced with scrim with a special UV protection.
- Covering material shall be non-metallic.
- Covering material shall provide mechanical resistance with an excellent aesthetic look.
- Covering material shall be mat finish.
- Covering material shall be of 380 ± 20 GSM as per ASTM D 646.
- Covering material shall be of 380 – 400 microns as per ASTM E 252
- Covering material shall have the tensile strength of >5 MPa in MD & CD as per ASTM D 638
- Covering material shall have good puncture and tear resistance.
- Covering material shall have good UV resistance as per ISO 4892-2 Method A tested for 1000 Hrs.

9.1.4 Acoustic Insulation Works (HVAC Internal Duct Lining)

Comply

- The acoustic insulation material shall be dust and fiber free open cell elastomeric nitrile rubber
- The acoustic insulation material shall have good sound absorption, damping and barrier properties.
- The density of acoustic insulation material shall be within 140-180 Kg/m3.
- The acoustic insulation material shall have EPA approved antimicrobial product protection and shall pass Fungi / Mold Resistance as per DIN EN ISO 846 method A, Bacterial Resistance as per DIN EN ISO 846 method C and Viral protection as per JIS Z 280 respectively, from an independent testing agency.
- The acoustic insulation material shall have a thermal conductivity not exceeding 0.047 W/(m·K) @ 20 Deg. C as per EN 12667.
- The acoustic insulation material shall withstand maximum surface temperature of +85 Deg. C and minimum surface temperature of -20 Deg. C as per EN 14706.
- The material shall conform to Class 1 rating for surface spread of Flame in accordance to BS 476 Part 7 & UL 94 (HB, V-0) in accordance to UL 94, 1996.
- The acoustic insulation material shall pass Air Erosion Resistance Test in accordance to ASTM Standard C 1071-05 without any laminate or covering (section 12.7), from an independent accredited testing agency.

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- Thickness of the insulation shall be as specified for the individual application.
- The insulation material shall be installed as per manufacturer's recommendation.

9.1.5 Adhesives for Insulation

- Special IGBC compliant for VOC adhesive for installing all flexible insulation materials Adhesive shall be used for bonding the seams securely together. All butt & longitudinal joints, fittings and coverings must be closed with adhesive.
- The adhesive shall be specially formulated for the Duct or Pipe insulation application and supplied by insulation manufacturer. The adhesive shall be free from benzene.
- The adhesive shall have excellent bonding to porous and non-porous surfaces.

Comply

5.7.14 For Duct Insulation - Technical Details

- Solvent Based Rubber contact adhesive
- Colour: Pale Yellow
Temperature Range: +5°C to +85°C
- Form: Liquid
- Material: Rubber based contact adhesive in blend of solvents
- Minimum Drying Time (airing time): 4-5 minutes
- Setting Time: 36 hours
- Viscosity @ 30°C: 700 - 1200 cps
- Specific Gravity: 0.80 - 0.85 g/cc
- Density: 0.083 kg/m³ + 0.01
- Flash Point: > 10 °C
- Coverage: 5-6 square meters of bonded area per Liter (Bonded area – Area of insulation material and duct together)
- Tack Retention Time: 25 minutes
- Insulation works to be done in a well-ventilated area with RH% below 80%

Comply

5.7.15 For Pipe, Underdeck & Wall Insulation - Technical Details

- Solvent Based Rubber contact adhesive
- Colour: Beige
Temperature Range: - 50°C to +105°C
- Form: Liquid
- Material: Contact adhesive on poly-chloroprene basis, free of aromatic solvent
- Aging Stability: Very good
- Resistance to weathering: Very good
- Working Time at 20 °C: Wet adhesion time: up to 2 minutes
- Minimum Drying Time (airing time): 3-5 minutes
- Contact Adhesion Time: 15-20 minutes
- Setting Time: 36 hours

9.1.6 Supply and Application Quality Check – Insulation materials

- All Third-party laboratories test certificates pertaining to each project specifications to be submitted by the manufacturer before material supply, and to be checked and verified by the PMC and Consultant before material supplies
- Supplied product brand name to be mentioned in the third-party test certificate.

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- Every supply should accompany quality confirmation certificates from manufacturer mentioning adherence to project specifications & batch/run number for traceability
- Each Insulation material should be printed with approved product's brand name and its respective batch/run number for traceability
- Dimension check – Supplied materials to be randomly checked for dimensions like thickness (by vernier caliper), width and length by Bidder in presence of PMC/User engineer. Insulation material should adhere to the project specifications dimensions. In case of mismatch, the complete supplied lot be hold and checked and manufacturer should rectify the issue and supply fresh materials with quality adherence.
- FM traceability – Product's brand name to be listed in FM website.
- Test of supplied materials to be conducted by PMC/User – material should be randomly selected from the supplies and sent for testing to the reputed labs having NABL accreditation. Parameters like k value, covering thickness, density, ASTM e85(25/50), class o and VOC (adhesive) to be tested. In case of mismatch, the complete supplied lot be hold and rejected and the cost of the test to be debited to the manufacturer and they should supply fresh materials with quality adherence. Cost of logistics for rejected and replacement materials to be borne by manufacturer.
- Application workmanship quality – Application should be done as per manufacturer guidebook or manual.
- Manufacturer representative should regularly visit the project site in a monthly basis and submit the observation report to PMC/User Engineering team. Any deviation in workmanship should be immediately raised and material should be reapplied in problematic area. Cost to be borne by the Bidder.

9.1.7 Handling, Storage & Housekeeping of Insulation

- Temperature – Store at room temperature.
- Avoid exposure to direct sunlight, artificial light, or UV
- Avoid exposure to any liquids or vapors
- Deformation: Avoid staking, Tension, bending, and twisting should be avoided
- Always keep insulation rolls in the horizontal direction.
- Stack only 5 layers of insulation over each other.
- Do not drag OR thump the roll during handling of rolls. It will cause damage to the rolls OR will cause wrinkles on the roll.
- Avoid keeping the adhesive brushes in the adhesive holder during application. Always keep a cover on the adhesive when not using. (A scrap section of Insulation sheet will do the job.)
- Keep adhesive brushes stored in thinners when not in use, this will keep the brushes clean during installation. (Remember to remove the thinner from the brushes before applying the adhesive to the insulation/ductwork services.)
- Avoid damaging the insulation during installation and fabrication.
- Keep the Insulation as clean as possible during fabrication. Where appropriate fabrication and adhesive work should be done off the ground.
- Keep Insulation roll inside the box during fabrication. Cut a “slot” out of the Insulation box, allowing for free movement of the Insulation sheet during fabrication.
- Use the correct size of knife and brush, for large surface areas the use of a paint roller may be the preferred option. (Avoid soft sponge roller sleeve types, better to use a short pile nylon type.)
- Brushes, use the following sizes for different applications as listed below;
- Surface of ductwork – large brush – 50mm to 100mm

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- Surface of Insulation sheet – metal sheet for spreading a thin and even layer, 25mm to 50mm brush for the edges of the Insulation sheet.
- Compressed seams and joints – long thin brush – 15mm to 20mm.
- Note: Always cut down the brush heads.
- After installation of Insulation clean the dust from all visual surfaces.
- Ensure the specified insulation sheet thickness is maintained throughout the installation, including all load support brackets.
- All ductwork surfaces should be cleaned before installing the insulation sheet where possible.
- Where possible, keep the insulation clean during fabrication.
- Plan out the proposed layout and dimensions of the ductwork, mark sizes with a marking pen, to speed up pre-fabrication production.

Insulation thickness values will be selected as per following selection table.

LT Chilled Water Pipe - 7 Deg. C		MT Chilled Water / Process Cooling Water Pipe - 13 Deg. C / Hot Water Pipe	
Dia. (NB)	Insulation Thickness(mm)	Dia. (NB)	Insulation Thickness(mm)
15	19	15	13
20 - 40	25	20 - 65	19
50-100	32	80 - 200	25
125 - 200	38	250 - 900	32
250 - 550	44	900 - 1200	38
600 - 1200	50		

Notes for Piping:

- Internal Chiller Plant Room Insulation shall be considered with Aluminum 24G Cladding
- From Plant Room to all other areas piping shall be considered with the Dura Clad.
- 15NB to 100NB Pipe Readymade tubes shall be used for Dura clad and Rest shall be considered with the Sheet Form.

Ducting insulation thickness shall be as per table below.

Ducting position	Insulation Thickness
SA duct – MAU / TFA	50mm for Plenum of Duct 32mm for First 15 meter of SA

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	Duct of MAU 25mm for SA Duct After 15 Meter
SA duct – AHU / RCU / Others	19mm
Return air system	13mm

9.2 PUMP INSULATION

Chilled water pump shall be insulated to the same thickness as the pipe to which they are connected and application shall be same as above. Care shall be taken to apply insulation in a manner as to allow the dismantling of pumps without damaging the insulation.

9.3 CHILLED WATER EXPANSION TANK INSULATION

Cold water tank, and chilled water expansion tank shall be insulated with insulated with 25mm Nitrile rubber and 26 g aluminum cladding.

9.4 MEASUREMENT OF INSULATION

Unless otherwise specified measurement for duct and pipe insulation for the project shall be based on center line measurements described herewith

- Pipe Insulation shall be measured in units of length along the center line of the installed pipe, strictly on the same basis as the piping measurements described earlier. The linear measurements shall be taken before the application of the insulation. It may be noted that for piping measurement, all valves, orifice plates and strainers are not separately measurable by their number and size. It is to be clearly understood that for the insulation measurements, all these accessories including cladding, valves, orifice plates and strainers shall be considered strictly by linear measurements along the center line of pipes and no special rate shall be applicable for insulation of any accessories, fixtures, or fittings whatsoever.
- Duct Insulation and Acoustic Lining shall be measured based on surface area along the center line of insulation thickness. Thus, the surface area of externally thermally insulated or acoustically lined be based on the perimeter comprising center line (of thickness of insulation) width and depth of the cross section of insulated or lined duct, multiplied by the center-line length including tapered pieces, bends, tees, branches, etc. as measured for bare ducting.

10. FAN FILTER UNITS:

10.1 SCOPE

The scope of this section includes the design, manufacture, supply, installation, testing and commissioning of Fan Filter Units (FFU) for final filtration of recirculated cleanroom air and delivery of filtered air to the Cleanroom FAB level, in accordance with project drawings, BOQ and these specifications.

The FFUs shall draw return air from the interstitial plenum and return air shafts, filter the air through high efficiency HEPA filters, and supply the filtered air into the cleanroom at a constant airflow rate to maintain the specified cleanroom classification.

Noted

The FFU system shall be designed to ensure: